

Secretary Problem Research

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Lecture 2: Relaxed SP

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1 Relaxed SP Prophet Inequality

1.1 classical SP PI

We have n items with weights $X_1, \dots, X_n \sim \text{Unif}(0, 1)$. We adopt a thresholding scheme for ALG: We pick the first item encountered with weight $\geq T$.

$$\begin{aligned} \mathbb{P}[X_{\max} \text{ selected}] &= \sum_{i=1}^n \mathbb{P}[X_1, \dots, X_{i-1} < T \cap X_i \geq T \cap X_i \text{ is max}] \\ &= \sum_{i=1}^n \int_0^1 \mathbb{P}[X_1, \dots, X_{i-1} < T \cap X_i \geq T \cap X_{i+1}, \dots, X_n \leq X_i \mid X_i = x] f_{\text{Unif}(0,1)}(x) dx \\ &= \sum_{i=1}^n \int_T^1 \mathbb{P}[X_1, \dots, X_{i-1} < T \cap X_{i+1}, \dots, X_n \leq x] f_{\text{Unif}(0,1)}(x) dx \\ &= \sum_{i=1}^n \int_T^1 T^{i-1} x^{n-i} dx \\ &= \sum_{i=1}^n \frac{1}{n-i+1} T^{i-1} (1 - T^{n-i+1}) \end{aligned}$$

1.2 relaxed model PI

1.2.1 calculation 1 (in meeting)

$$\begin{aligned}
& \sum_{i=1}^n \mathbb{P}[\text{accept } i \cap i \text{ best}] \\
&= \sum_{i=1}^n \mathbb{P}[\text{accept } i \mid i \text{ best}] \mathbb{P}[i \text{ best}] \\
&= \frac{1}{n} \sum_{i=1}^n \mathbb{P}[X_1, \dots, X_{i-1} < T \cap X_i \geq T \mid i \text{ best}] \\
&= \frac{1}{n} \sum_{i=1}^n \mathbb{P}[X_1, \dots, X_{i-1} < T \mid X_i \geq T \cap i \text{ best}] \mathbb{P}[X_i \geq T \mid i \text{ is best}]
\end{aligned}$$

Note that $\mathbb{P}[X_i \geq T \mid i \text{ best}] = 1 - F_{X_{\max}}(T) = 1 - T^n$. Also,

$$\begin{aligned}
& \mathbb{P}[X_1, \dots, X_{i-1} < T \mid X_i \geq T \cap i \text{ best}] \\
&= \int_T^1 \mathbb{P}[X_1, \dots, X_{i-1} < T \mid i \text{ best} \cap X_i = x] f_X(x \mid X_i \geq T \cap i \text{ best}) dx \\
&= \int_T^1 \mathbb{P}[X_1, \dots, X_{i-1} < T \mid X_1, \dots, X_{i-1} < x] f_{X_{\max}}(x \mid X_{\max} \geq T) dx \\
&= \int_T^1 \left(\frac{T}{x}\right)^{i-1} \frac{nx^{n-1}}{1-T^n} dx \\
&= \frac{nT^{i-1}}{1-T^n} \int_T^1 x^{n-i} dx \\
&= \frac{nT^{i-1}}{1-T^n} \frac{1-T^{n-i+1}}{n-i+1}
\end{aligned}$$

Combining all terms,

$$\begin{aligned}
& \sum_{i=1}^n \mathbb{P}[\text{accept } i \cap i \text{ best}] \\
&= \frac{1}{n} \sum_{i=1}^n \frac{nT^{i-1}}{1-T^n} \frac{x^{n-i+1}}{n-i+1} (1-T^n) \\
&= \sum_{i=1}^n T^{i-1} \frac{1-T^{n-i+1}}{n-i+1}
\end{aligned}$$

1.2.2 Calculation 2

We have n items with weights $X_1, \dots, X_n \sim \text{Unif}(0, 1)$. Observed values are $\tilde{X}_i \sim \text{Unif}(X_i - \delta, X_i + \delta)$. We adopt a thresholding scheme for ALG: We examine an item if $\tilde{x}_i > T$ and accept if $x_i > T$.

$$\begin{aligned}
 \mathbb{P}[X_{\max} \text{ selected}] &= \sum_{i=1}^n \mathbb{P}[X_1, \dots, X_{i-1} < T \cap X_i \geq T \cap X_i \text{ is max}] \\
 &= \sum_{i=1}^n \int_0^1 \mathbb{P}[X_1, \dots, X_{i-1} < T \cap X_i \geq T \cap X_{i+1}, \dots, X_n \leq X_i \mid X_i = x] f_{\text{Unif}(0,1)}(x) dx \\
 &= \sum_{i=1}^n \int_T^1 \mathbb{P}[X_1, \dots, X_{i-1} < T \cap X_{i+1}, \dots, X_n \leq x] f_{\text{Unif}(0,1)}(x) dx \\
 &= \sum_{i=1}^n \int_T^1 T^{i-1} x^{n-i} dx \\
 &= \sum_{i=1}^n \frac{1}{n-i+1} T^{i-1} (1 - T^{n-i+1})
 \end{aligned}$$

Lecture 3: Relaxed SP with error

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2 Calculation with error involved

The weights $X_1, \dots, X_n \sim \text{Unif}(0, 1)$. The noisy observations $\tilde{X}_1, \dots, \tilde{X}_n \sim \text{Unif}(x_i - \delta, x_i + \delta)$.

The strategy is to introduce thresholding parameters R and $T \in (0, 1)$. We examine the upcoming item i if $\tilde{x}_i > R$, and accept the item if $x_i > T$. We still want to find $\mathbb{P}[X_{\max} \text{ is selected}]$ given particular R, T, n, δ .

$$\mathbb{P}[X_{\max} \text{ selected}] = \sum_i \int_0^1 \mathbb{P}[X_i \text{ best} \cap i \text{ selected} \mid X_i = x] f(x) dx \quad (1)$$

$$\begin{aligned}
 &\mathbb{P}[X_i \text{ best} \cap i \text{ selected} \mid X_i = x] \\
 &= \mathbb{P}[\text{reach } i \cap \tilde{X}_i > R \cap X_i > T \cap X_i \text{ max} \mid X_i = x] \\
 &= \mathbb{P}[\text{Unif}(x - \delta, x + \delta) > R] \mathbf{1}(x > T) \mathbb{P}[\text{reach } i \cap X_i \text{ max} \mid X_i = x]
 \end{aligned}$$

$$\begin{aligned}
 & \mathbb{P}[\text{reach } i \cap X_i \text{ max} \mid X_i = x] \\
 &= \sum_{k=0}^{\lfloor \frac{i-1}{2} \rfloor} \mathbb{P}[\text{reach } i \cap X_i \text{ max} \cap k \text{ examines before reaching } i \mid X_i = x] \\
 &= \sum_{k=0}^{\lfloor \frac{i-1}{2} \rfloor} \mathbb{P}[\text{reach } i \text{ with } k \text{ examines}] \mathbb{P}[X_i \text{ max} \mid \text{reach } i \text{ with } k \text{ examines} \cap X_i = x] \\
 &= \sum_{k=0}^{\lfloor \frac{i-1}{2} \rfloor} \underbrace{\frac{(i-1-k)!}{k!(i-1-2k)!}}_{\text{ways to reach } i} \mathbb{P}[\text{ED}]^k \mathbb{P}[\text{D}]^{i-1-2k} \underbrace{F(x|X < T)^k}_{\text{examined}} \underbrace{F(x)^k}_{\text{omitted}} \underbrace{F(x|\tilde{X} < R)^{i-1-2k}}_{\text{declined directly}} \underbrace{F(x)^{n-i}}_{\text{upcoming}}
 \end{aligned}$$

where

$$\begin{aligned}
 \mathbb{P}[\text{ED}] &= \frac{1}{2\delta} \int_0^T \int_{y-\delta}^{y+\delta} \mathbf{1}(z > R) dz dy = \text{const} \\
 \mathbb{P}[\text{D}] &= \frac{1}{2\delta} \int_0^1 \int_{y-\delta}^{y+\delta} \mathbf{1}(z < R) dz dy = \text{const}
 \end{aligned}$$

and for $0 \leq x \leq 1$:

$$\begin{aligned}
 F(x|X < T) &= \begin{cases} \frac{x}{T} & 0 \leq x < T \\ 1 & x \geq T \end{cases} \\
 F(x|\tilde{X} < R) &= \frac{\int_0^x \int_{y-\delta}^{y+\delta} \mathbf{1}(z < R) dz dy}{\int_0^1 \int_{y-\delta}^{y+\delta} \mathbf{1}(z < R) dz dy} \\
 F(x) &= x.
 \end{aligned}$$